

TECHNICAL BRIEF

Solar PV Production Forecasting Using Machine Learning

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THE PROBLEM

Solar energy production is inherently variable. Weather, cloud cover, and seasonal shifts create uncertainty that makes it difficult for operators to optimize workload scheduling, battery storage, and grid interactions. Traditional solar forecasting can be improved by adding in real time cloud cover forecasts. All this helps a local microgrid to maintain operation. Without accurate short-term forecasts, solar-powered operations leave significant efficiency gains on the table.

APPROACH

Build and compare supervised learning models to predict hourly PV output from weather and temporal features:

Model	Role
Ridge / Lasso	Regularized baseline
Random Forest	Non-linear ensemble
XGBoost	Gradient boosting (primary)
LSTM	Temporal sequence (stretch)

KEY FEATURES

Weather	Engineered
GHI / DNI / DHI Air Temperature Wind Speed Humidity Cloud Cover	Hour (sin/cos) Day of Year (sin/cos) Lag features (1h, 24h) Rolling means (3h, 6h) GHI × Temp interaction

EXPECTED OUTCOMES

- XGBoost to outperform linear baselines by 15–25% RMSE reduction
- GHI and cloud cover as dominant predictors of output variability
- Lag features to provide significant accuracy gains over weather-only models

EVALUATION

- RMSE and MAE for prediction accuracy
- R² for variance explained
- Time-series cross-validation (expanding window)
- Feature importance ranking across models

DATA SOURCES

PVDAQ (Photovoltaic Data Acquisition)	An NREL-managed DOE database of real measured PV performance data. Primary site: System 4902 (NIST Ground-1, 270.7 kW fixed ground-mount, Gaithersburg MD), 98 sensor channels at 1-min resolution, 3.5 years. Includes POA irradiance, temperature, wind, AC/DC power.
NSRDB (National Solar Radiation Database)	An NREL-maintained database of hourly solar radiation and weather data. Supplements PVDAQ for cross-site generalization testing with System 1332 (NREL Parking Garage, Golden CO, 1,153 kW, 11+ years, power only). Supplementary hourly weather data (GHI, DNI, DHI, temperature, wind, humidity)

PRACTICAL APPLICATION

These forecasts enable dynamic workload scheduling for solar-powered compute infrastructure—ramping GPU clusters, mining rigs, and AI inference nodes during high-production windows while optimizing battery charge/discharge cycles and EV charging during predicted surplus periods. The model’s feature importance analysis will guide sensor deployment priorities for future installations, and the forecasting pipeline is designed to integrate directly into real-time asset management platforms for automated alerting and scheduling.

GLOSSARY OF TERMS

Key concepts and terminology referenced in this technical brief

SOLAR RADIATION & ENERGY

GHI (Global Horizontal Irradiance)	Total amount of solar radiation received on a horizontal surface, measured in watts per square meter (W/m ²). Includes both direct sunlight and diffuse sky radiation. This is the most commonly used measure for estimating solar panel output.
DNI (Direct Normal Irradiance)	Solar radiation received directly from the sun on a surface perpendicular to the sun's rays (W/m ²). Important for tracking solar panel systems that follow the sun's position throughout the day.
DHI (Diffuse Horizontal Irradiance)	Solar radiation scattered by the atmosphere (clouds, dust, water vapor) that reaches a horizontal surface indirectly (W/m ²). On cloudy days, DHI makes up most of the available solar energy.
PV (Photovoltaic)	Technology that converts sunlight directly into electricity using semiconductor materials. Solar panels are made up of many PV cells wired together.
kW / kWh	Kilowatt (kW) measures power output at a point in time. Kilowatt-hour (kWh) measures total energy produced over time. A 5 kW solar system running at full capacity for 1 hour produces 5 kWh of energy.

MACHINE LEARNING MODELS

Ridge / Lasso Regression	Linear models that add a penalty to prevent overfitting. Ridge (L2) shrinks all feature weights evenly; Lasso (L1) can zero out less important features entirely, effectively selecting the most relevant inputs. Used here as an interpretable baseline.
Random Forest	An ensemble method that builds many decision trees on random subsets of the data and averages their predictions. Handles non-linear relationships well and provides built-in feature importance rankings.
XGBoost (Extreme Gradient Boosting)	An advanced tree-based algorithm that builds models sequentially, with each new tree correcting errors from the previous ones. Currently one of the most effective algorithms for structured/tabular data prediction tasks.
LSTM (Long Short-Term Memory)	A type of neural network designed for sequential data. It can learn long-range dependencies in time series—for example, recognizing that weather patterns from 24 hours ago affect today's solar output. Listed as a stretch goal due to added complexity.

EVALUATION METRICS

RMSE (Root Mean Squared Error)	Measures average prediction error in the same units as the target variable (e.g., kW). Larger errors are penalized more heavily. Lower RMSE = better predictions. This is our primary accuracy metric.
MAE (Mean Absolute Error)	Average of the absolute differences between predicted and actual values. Less sensitive to occasional large errors than RMSE. Gives a straightforward "average miss" in real units.
R² (R-Squared)	Proportion of variance in the target variable explained by the model, ranging from 0 to 1. An R ² of 0.85 means the model explains 85% of the variability in solar output. Higher is better.
Cross-Validation (Expanding Window)	A method for testing model reliability on time series data. The training set grows over time while the test set always comes after—mimicking how the model would be used in practice, where you only have past data to predict the future.

FEATURE ENGINEERING CONCEPTS

Cyclical Encoding (sin/cos)	A technique for representing repeating patterns (like hour of day or season) as smooth curves rather than arbitrary numbers. Hour 23 and hour 0 are close together in reality—sin/cos encoding captures this, while raw numbers don't.
Lag Features	Previous time period values used as inputs. A "1-hour lag" of solar output means using last hour's production to help predict this hour's. Captures momentum and local atmospheric persistence.
Rolling Means	Averages computed over a sliding time window (e.g., the last 3 or 6 hours). Smooths out short-term noise and captures recent trends in weather or production patterns.
Interaction Features	New variables created by combining two features (e.g., GHI × Temperature). Captures relationships where the effect of one variable depends on another—for example, high irradiance at extreme temperatures may reduce panel efficiency.